

My Design intent

CSD2541

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Introduction

My name is Xin, an aspiring level designer focusing on creating clear and intentional player experiences through structuring each level and properly defining how mechanics are used. To me, level design goes beyond simply making a space that fits requirements, it is about putting myself into the player's shoes, thinking about how they would go through each part, what they should be doing, what they should learn and how they should feel throughout each section. Doing this allows me to fully map out how the level itself should play out.

My process usually begins from one of two starting points:

Theme-Led Design

- Identity, background and function guides how the level itself is designed.
- Essentially, the “story” of the world is decided first, then the level is designed to fit it.



Fig 1.

Map with a “Solo-Pirate” and “Self-sustaining” theme

Mechanics-Led Design

- Abilities, world features and their interactions shape the level.
- How the level should be played is planned out first, then the level is designed to facilitate the experience.



Fig 2.

Map designed for “mid-air dash” and “double-jump”, mobility-based abilities

I choose between them mostly depending on what the level is actually intended for, and what constraints are given, as well as what available systems I have. For example, a level that is mostly used for traversal, safe-zones, lobbies, story-telling and such will typically use Theme-Led, while a level that is meant to be more of a skill-check, a challenge, or a puzzle will typically use Mechanics-Led.

While different at the start, both approaches rely on the same techniques of **planning**, **evaluation** and **iteration**.

To do so, I utilise various tools and methods, such as **programme planning**, **skill themes** and **objectives** to organise my initial idea and get a better picture of how to approach. then refining the finer details through deciding the appropriate **scale and metrics**, **composition hierachy** and **map circulation** just to name a few. The idea is then tested by myself and other playtesters before I update and/or balance the level via **iteration**, changing the **engagement beats** , **pacing curve** and **resource distribution**, especially on the **golden path**.

The following pages will explain in more detail how I utilise these methods, and show cases where I applied them. I will also be evaluating some of my work, the process I used to complete it, and what I learnt from that process, as well as how I've used what I learnt to improve upon my process.

Turning an Idea into a level

Before jumping into any designing, I first gather all available information about said level, and gauge the degree of freedom I have to work with. I find out what elements are already predetermined for me, and what I get to personally decide. The function of level would be the most important, and if it was predetermined, would greatly impact which of my two directions I take. Other factors such as scale, style, etc are also important.

1.Programme

With all that information under my belt, I then start to map out the intended player experience, creating and testing out some ideas. I would typically do so via **programme**, where I can create a barebones version of the map quickly to see clearly how the maps would connect and interact with each other. Through this, I can better visualise how the map would look like, and can decide from there if that is the direction I want to go with. With this, I can also better see if the **scale and metrics** of the map fits my requirements, as well plan out the **map circulation**, seeing how movement is between each rooms, whether there is backtracking, restrictions, etc.

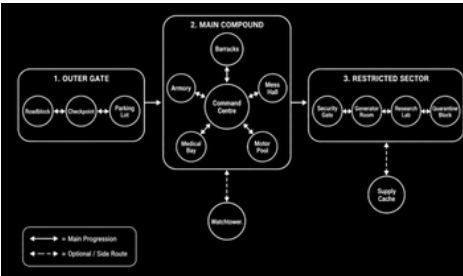


Fig 3.

- Initial programme for my zombie apocalypse survival military base map
- Rejected as I did not think it was sufficient for my idea

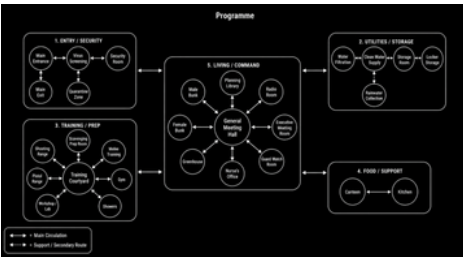


Fig 4.

- Programme for my school repurposed to survival base map for zombie apocalypse
- Proceeded as I liked the structure of this map

2.Skill themes/objectives

Once the structure is decided, I start to plan out the experience itself. Depending on the type of level I'm making, I would do this via **skill themes and/or objectives**. Both picking a skill theme and creating objectives accomplishes a similar goal, it streamlines the designing process. This is because instead of struggling due to a lack of direction, these tools can act as a guide for us to follow and eventually converge onto a proper idea. We can also utilise these to decide which spaces are likely more important(by gauging its relevant to the objective/skill), thus emphasising it and guiding players properly via **composition hierachy**, which attempts to use human's different innate biases to control their point of focus.

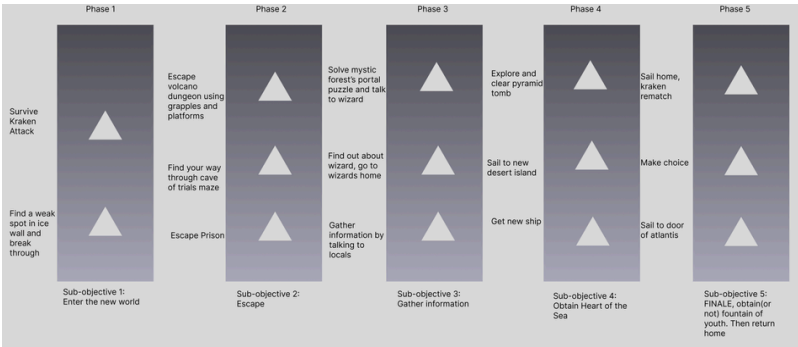


Fig 5.

My objectives and key interaction events for my 150x150 map
Creating this helped me split up my map into different sections for different parts of the game. Greatly helped organisation and thought process

#	Event	Details	Duration	Intensity	Type	Extra
1	Breach the Ice Wall	Jacob investigates the Ice Wall, finds a weak point, and uses the ship's cannons to break through.	5 min	3	Main	Introduces exploration, ship control, and cannon mechanics.
1.1	Frozen Treasure	Jacob discovers treasure trapped inside the Ice Wall.	3 min	2	Side	Optional exploration reward.
1.2	Shipwreck	Jacob salvages nearby shipwrecks.	3 min	3	Side	Optional salvage quest. Intensity increase due to seeing shipwreck.
2	Kraken Attack	A giant Kraken attacks shortly after Jacob enters the New World. Fight back!	8 min	8	Main	Initial spike in intensity. Winning gives Jacob some kraken parts that is worth gold.
3	Shipwrecked	Jacob's ship is destroyed, forcing him to wash ashore on an island. Salvage as much loot as possible.	3 min	5	Main	Transitions from ship gameplay to survival gameplay.
4	Captured	Local islanders capture Jacob and imprison him inside a volcano settlement.	1 min	5	Main	Ends Phase 1 and creates the next sub-goal.

Fig 6.

Events table for the first section of my 150x150 map. This helped me to plan the specifics required in this section of the map.

3.Pacing curve/Golden Path

Once the map itself is created and populated, I start thinking of how the **pacing curve** should go, as well as what the **golden path**(ideal route players can take) should look like. I typically like to start my level off with either an initial spike of intensity to hook the players in (**theme-led design**), or a tutorial-like area to slowly ease the players into the game(**mechanic-led design**).

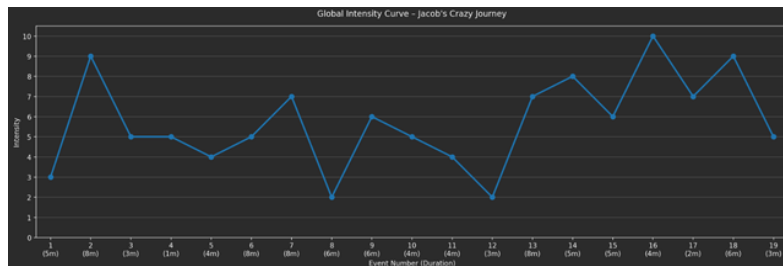


Fig 7.

Initial spike in intensity for my 150x150 level, a story-focused game

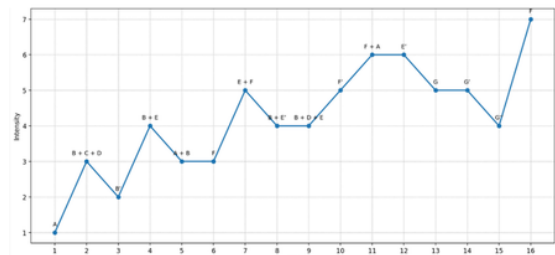


Fig 8.

Pacing curve for my prison-break megaman stage, easing players in, gradual increase from start to end

Either way, after the starting area, I would gradually increase the intensity of the game as player's go through it, having an good amount of low intensity "down-times" accompanied by the jump in intensity during an crucial part, with the jump increasing as the game proceeds either due to the stakes of the story increasing or player skill proficiency increasing. This is done in order to keep player at an optimal level of arousal, also known as engagement, in order to have the player in the "flow-state" while playing.

During this process, I would likely go back to previous steps to make slight edits to fit my final vision more, such as changing the order of some objectives and changing how often meaningful interactions occur(**engagement beats**). This is also where I decide my **resource distribution**, ensuring that enough resources is given out on the golden path itself, but also that there are an appropriate amount given in side quests and such to incentive players to explore outside of the main quest/route.

4.Improvement/feedback

At this point, I would have finished my first iteration of the level. From here, I will look at the complete map again to see if there is anything I'm personally unsatisfied with and would like to change. This is also the stage where I would attempt to ask my colleagues or the person in charge for some feedback, getting a second opinion on the matter. More often than not, I find the second opinion very impactful as they tend to be able to spot the imperfections that I miss, allowing me to further improve on the design. I will keep sending out my level to others to gather feedback and iterate on it until I am satisfied with how it is.

5.Final checks and reflection

Before finalizing the design, I do a last playthrough to ensure that user flow is natural and that edge cases are accounted for.

Upon finalization, I reflect on the entire design process, pointing out the good and the bad, and how I should improve in the future. This helps me polish the parts im good at while improving those I'm weaker at.

I make it a point to not reiterate on completed designs to ensure that I have a better view on my progress made as a level designer. Being able to look back at my past work to see how my design methodologies have changed, helps me to improve more consistently.

My design philosophy

With my main design method now clear, I will go on to explain my design philosophies. What is the difference you may ask? Well, personally, my design methods is sort of like a personal formula that I believe in, as well as continuously improve on, something that I can use as an outline for any level design. My design philosophy, on the other hand, is how I approach step of these steps. It is how I perceive “good level design”.

1.Difficulty VS Complexity

As someone who is a big fan of notoriously difficult games such as “Dark Souls”, I highly value difficulty as a factor in level design. This is not to be confused with complexity. In my opinion, games bloated with tons of mechanics and features are just compensating for inferior level design. Clean gameplay focusing on just a couple of mechanics at a time, while steadily increasing the difficulty and required mastery of those mechanics, that is what I believe a good level should accomplish. Players should not have to be overwhelmed by the amount of different features available at once, they should be able to have a clear idea of what to do with what tools.

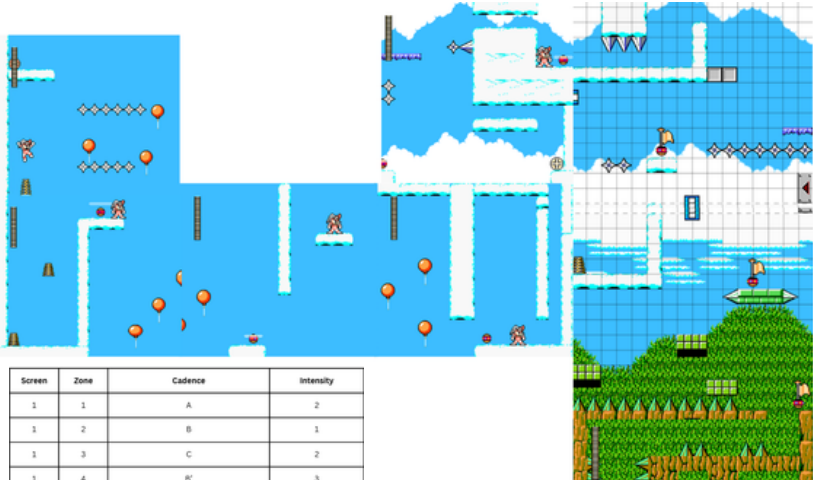


Fig 9. JumpMan level + Fig 10. JumpMan cadence

- On the left is a megaman level I created, “JumpMan” along with its cadence
- This level applies skill themes, with it’s theme being movement based skills throughout the entire map.
- There are **TWO** primary mechanics, “Double-Jump” and “Mid-Air Dash”, both of which are mobility based skills, and **ONE** secondary mechanic, “Timing”, where players have to time their jump to fit various different obstacles such as disappearing floors.
- Attached below is my **cadence notation** for this level. For reference, A is jump, B is double jump and C is dash. As you can see, throughout the entire level, every single challenge has at least one of the three, often times multiple.

Screen	Zone	Cadence	Intensity
1	1	A	2
1	2	B	1
1	3	C	2
1	4	B'	3
1	5	A+C	5
1	6	B	1
2	7	B+C+D	4
2	8	B'+C'	6
2	9	A'+C'	3
3	10	B+C+D+E	7
3	11	C	2
4	12	B'+C'	6
4	13	B+E OR C+E	4
4	14	B+C'	6
5	15	B+C+D	5
6	16	C'	4
7	17	B+C+D'	7
7-8	18	A+B+C+C'+D'	9

What I did here was choose the main mechanic I wish to have for this level, and focus on them, gradually increasing the difficulty by requiring higher skill with the same mechanic for later challenges. For example, my earlier challenges only required players to activate mid-air dash to complete, whereas later challenges require precise timing of the dash in order to dash through a one-block gap midair.

2.Guided freedom

Another key element of level design I believe in is freedom, but with direction. I’ve played many open world games, and the one thing I always hate is not knowing what to do next. There always seems to be too many things to do, overwhelming me and lessening my enjoyment of the game. With that in mind, I come to the concept of “guided freedom”. I believe that for a level to be good, players SHOULD be given the freedom to do whatever they want, but this freedom should almost be like an “illusion”. Whether by incentives, punishments or other methods, I believe in technically giving players freedom but yet obviously having a “correct” path. This reduces frustration of players getting lost in the game while also satiating players who want more agency.



Fig 11. Section 3 of ONE million campaign

#	Event	Details	Duration	Intensity	Type	Extra
8	Main Island Exploration	Jacob explores the main settlement, speaks with townsfolk, and gathers rumors about the Fountain. Learns about the reward in the mystic forest.	6 min		2 Main	Recovery phase after the volcano.
8.1	Ghost Chef	Jacob helps deal with ghost chefs at the local restaurant.	4 min		1 Side	extra side quest
8.2	Farmer's back issues	Jacob helps a farmer with his crops.	5 min		1 Side	adds some realistic side quest
8.3	Blacksmith	Jacob helps gather ores for the blacksmith from the mines.	5 min		2 Side	Optional combat activity.
8.4	Fishing Tournament	Jacob joins a strange magical fishing competition.	5 min		3 Side	Optional low-intensity activity.
8.5	Bounty Board	Jacob hunts wanted pirates around the island.	5 min		5 Side	Optional combat and exploration.
9	Mystic Forest	Jacob enters the magical forest and navigates weird teleporting circles.	6 min		6 Main	Introduces magical navigation puzzles.
9.1	Lost Explorer	Jacob rescues an adventurer trapped in the forest.	4 min		2 Side	Optional rescue quest.
9.2	Dinosaur encounter	While navigating the forest, Jacob runs into an ancient creature. Fight to survive!	5 min		7 Side	Optional Boss
10	Wizard's Wisdom	Upon passing the wizard's trial and reaching the reward, Jacob learns from the wizard the location of the fountain, and the	4 min		5 Main	Important story reveal. Location: Atlantis King: Heart of the Sea

Fig 12. Events table for section 3 of ONE million campaign

The above is a portion of my 150x150 campaign map, as well as its events table on the right. In this section of the game, our main character “Jacob”, enters through the top left of the map. As we can see in the events table, there are a ton of side quest. This is what I would consider the hub town of the game. Despite having a ton of different things to do now, the player is given two clear goals; **Earn money for a new ship AND obtain information on the fountain of youth**

This gives players the feeling of freedom as they can choose to do whatever they want, but still giving them direction as they know in the back of their head what they eventually are working towards. On top of that, upon getting a new ship, I placed hard bosses on parts of the ocean that isn't on track of the main quest, both as a warning to dissuade players from going there AND to reward those who do attempt this anyways, satisfying both types of players.

3.No idea is too crazy!

Lastly, and perhaps the one that I hold dearest to my heart, the believe that no idea is too crazy. I always didn't like having rigid rules, giving myself any strict restrictions, and this applies to my level design as well. Who is to decide whether an idea is insane or crazy? Isn't the point of games to be able to create an experience that wouldn't be possible in other mediums? With that in mind, I attempt to try to come up with unique and even somewhat “bizarre” ideas to work with. I really enjoy trying to make something that doesn't normally seem like it would work, work.



Fig 13. Dinosaur scientist + Assistant(WFH2)

- Mad scientist reviving dinosaurs.
- Scientist can't do basic chores, require assistant's help to survive



Fig 13. Goat Lover(WFH3)

- Man is obsessed with goats.
- Built his house in the shape of a goat
- Structure his house entirely around the needs of goats

Conclusion

My approach to level design pairs structured planning in terms of my design methods with creativity and experimentation in terms of my design philosophies. Whether for thematic use or a challenging level, I have tools and methods to use to design them, such as programmes, pacing curves, etc. And as a level designer, I am constantly reflecting upon my own work, refining my methods and philosophy. I aim to continue to do refine these processes while creating experiences that are both novel and enjoyable for my players.